THE EFFECTS OF INFLATION AND DEFLATION ON INVENTORIES, COSTS OF SALES, AND GROSS MARGIN

Learning Outcome

- calculate and explain how inflation and deflation of inventory costs affect the financial statements and ratios of companies that use different inventory valuation methods

The allocation of the total cost of goods available for sale to cost of sales on the income statement and to ending inventory on the balance sheet varies under the different inventory valuation methods. In an environment of declining inventory unit costs and constant or increasing inventory quantities, first-in, first-out (FIFO) (in comparison with weighted average cost or LIFO) will allocate a higher amount of the total cost of goods available for sale to cost of sales on the income statement and a lower amount to ending inventory on the balance sheet. Accordingly, because cost of sales will be higher under FIFO, a company's gross profit, operating profit, and income before taxes will be lower.

Conversely, in an environment of rising inventory unit costs and constant or increasing inventory quantities, FIFO (in comparison with weighted average cost or LIFO) will allocate a lower amount of the total cost of goods available for sale to cost of sales on the income statement and a higher amount to ending inventory on the balance sheet. Accordingly, because cost of sales will be lower under FIFO, a company's gross profit, operating profit, and income before taxes will be higher.

The carrying amount of inventories under FIFO will more closely reflect current replacement values because inventories are assumed to consist of the most recently purchased items. The cost of sales under LIFO will more closely reflect current replacement value. LIFO ending inventory amounts typically are not reflective of current replacement value because the ending inventory is assumed to be the oldest inventory and costs are allocated accordingly. Example 3 illustrates the different results obtained by using either the FIFO or LIFO methods to account for inventory.

EXAMPLE 3

Impact of Inflation Using LIFO Compared to FIFO

Company L and Company F are identical in all respects except that Company L uses the LIFO method and Company F uses the FIFO method. Each company has been in business for five years and maintains a base inventory of 2,000 units each year. Each year, except the first year, the number of units purchased equaled the number of units sold. Over the five year period, unit sales increased 10 percent each year and the unit purchase and selling prices increased at the beginning of each year to reflect inflation of 4 percent per year. In the first year, 20,000 units were sold at a price of USD15.00 per unit and the unit purchase price was USD8.00.

1. What was the end-of-year inventory, sales, cost of sales, and gross profit for each company for each of the five years?

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Solution:

Company L using LIFO (in USD)	Year 1	Year 2	Year 3	Year 4	Year 5
Ending inventory[a]	16,000	16,000	16,000	16,000	16,000

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Sales[b]	300,000	343,200	392,621	449,158	513,837
Cost of sales[c]	160,000	183,040	209,398	239,551	274,046
Gross profit	140,000	160,160	183,223	209,607	239,791

[a] Inventory is unchanged at USD16,000 each year (2,000 units × USD8). 2,000 of the units acquired in the first year are assumed to remain in inventory.

[b] Sales Year X = (20,000 × USD15)$(1.10)^{X-1}(1.04)^{X-1}$. The quantity sold increases by 10 percent each year and the selling price increases by 4 percent each year.

[c] Cost of sales Year X = (20,000 × USD8)$(1.10)^{X-1}(1.04)^{X-1}$. In Year 1, 20,000 units are sold with a cost of USD8. In subsequent years, the number of units purchased equals the number of units sold and the units sold are assumed to be those purchased in the year. The quantity purchased increases by 10 percent each year and the purchase price increases by 4 percent each year.

If the company sold more units than it purchased in a year, inventory would decrease. This is referred to as LIFO liquidation. The cost of sales of the units sold in excess of those purchased would reflect the inventory carrying amount. In this example, each unit sold in excess of those purchased would have a cost of sales of USD8 and a higher gross profit.

Company F using FIFO (in US dollars)	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Ending inventory[a]	16,000	16,640	17,306	17,998	18,718
Sales[b]	300,000	343,200	392,621	449,158	513,837
Cost of sales[c]	160,000	182,400	208,732	238,859	273,326
Gross profit	140,000	160,800	183,889	210,299	240,511

[a] Ending Inventory Year X = 2,000 units × Cost in Year X = 2,000 units [USD8 × $(1.04)^{X-1}$]; 2,000 units of the units acquired in Year X are assumed to remain in inventory.

[b] Sales Year X = (20,000 x USD15)$(1.10)^{X-1}(1.04)^{X-1}$.

[c] Cost of sales Year 1 = USD160,000 (= 20,000 units × USD8). There was no beginning inventory.

Cost of sales Year X (where X ≠ 1) = Beginning inventory plus purchases less ending inventory

= (Inventory at Year X–1) + [(20,000 × USD8)$(1.10)^{X-1}(1.04)^{X-1}$] − (Inventory at Year X)

= 2,000(USD8)$(1.04)^{X-2}$ + [(20,000 × USD8)$(1.10)^{X-1}(1.04)^{X-1}$] – [2,000 (USD8)$(1.04)^{X-1}$].

For example, cost of sales Year 2 = 2,000(USD8) + [(20,000 x USD8) (1.10)(1.04)] – [2,000(USD8)(1.04)]

= USD16,000 + USD183,040 – USD16,640 = USD182,400.

2. Compare the inventory turnover ratios (based on ending inventory carrying amounts) and gross profit margins over the five-year period and between companies.

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Solution:

	Company L					**Company F**				
Year	**1**	**2**	**3**	**4**	**5**	**1**	**2**	**3**	**4**	**5**
Inventory turnover	10.0	11.4	13.1	15.0	17.1	10.0	11.0	12.1	13.3	14.6
Gross profit margin (%)	46.7	46.7	46.7	46.7	46.7	46.7	46.9	46.8	46.8	46.8

Inventory turnover ratio = Cost of sales ÷ Ending inventory.

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unchanged. The increase in the inventory turnover ratio is higher for Company L because Company L's cost of sales is increasing for inflation, but the inventory carrying amount is unaffected by inflation. It might appear that a company using the LIFO method manages its inventory more effectively, but this is deceptive. Both companies have identical quantities and prices of purchases and sales and only differ in the inventory valuation method used.

Gross profit margin = Gross profit ÷ Sales.

The gross profit margin is stable under LIFO because both sales and cost of sales increase at the same rate of inflation. The gross profit margin is slightly higher under the FIFO method after the first year because a proportion of the cost of sales reflects an older purchase price.

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Example # 3

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MO Something seems off in Year 2 with FIFO Company F. We assume that the number of units purchased equaled the number of units sold in subsequent years. However, we have 2,000 units left over from Year 1 @ $8/unit, with FIFO method, we have to account for those first, i.e., Year 2 COGS = 2,000 * $8 + 20,000 * $8 * 1.04 = $182,400, however, what they're doing is taking the entire 22,000 units at $8 * 1.04, which is incorrect, 2,000 of those units were purchased in Year 1 @ $8/unit. FIFO means the oldest inventory is used first. Am I missing something?

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AK The practice question for this question is new and has not been discussed before. This section is awful.

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